

Hyper-personalizing Common Norms through Principled Bespoke Generation

Jungeun Lee
POSTECH

Pohang, Gyeongbuk, South Korea
jelee@postech.ac.kr

Inseok Hwang
POSTECH

Pohang, Gyeongbuk, South Korea
i.hwang@postech.ac.kr

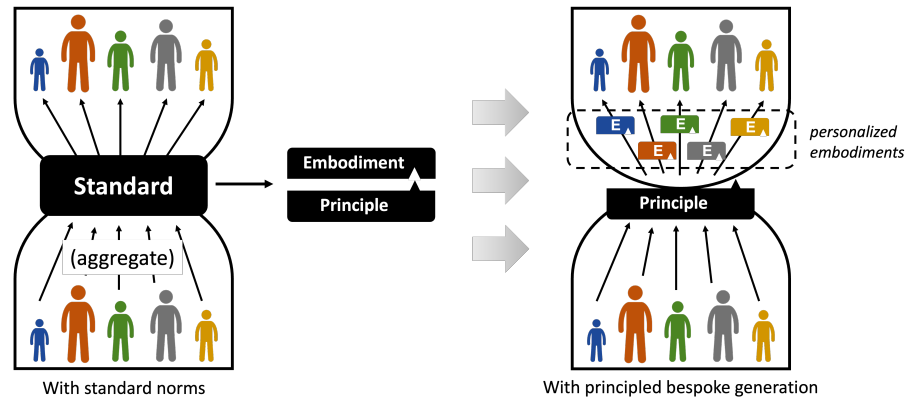


Figure 1: Overview of the philosophical aim of the thesis

ABSTRACT

In our everyday lives, we often follow standardized norms based on the “standard person”. However, these existing norms, created by averaging data from large populations, often fail to fit an individual. It is also challenging to implement guidelines based on these norms due to various real-life constraints in our life. We propose a *hyper-personalization* approach, leveraging principled generation to overcome these limitations. We explore a method that deconstructs the standard norms into principles and embodiments, and then generates contextualized guidelines by producing embodiments based on these principles. We initially identified the efficacy of our research by designing a real-life actionable sleep recommendation system. Our next step is the personalization of vocabulary in children’s language learning. In the end, we aim to design and implement a framework that can be applicable to potential cases in the future.

CCS CONCEPTS

• Human-centered computing → Ubiquitous computing.

KEYWORDS

personalization, sleep recommendation, language acquisition, principled generation

ACM Reference Format:

Jungeun Lee and Inseok Hwang. 2023. Hyper-personalizing Common Norms through Principled Bespoke Generation. In *Adjunct Proceedings of the 2023 ACM International Joint Conference on Pervasive and Ubiquitous Computing & the 2023 ACM International Symposium on Wearable Computing (UbiComp/ISWC ’23 Adjunct)*, October 08–12, 2023, Cancun, Quintana Roo, Mexico. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3594739.3610761>

1 PROBLEM STATEMENT

We encounter numerous standard guidelines in our daily lives. These range from basic health recommendations, such as “a man between ages 26-45 needs 2000-3000 calories per day” [1] to more intricate ones like “a 2-year-old child should know more than 50 words” [4]. These guidelines are established to evaluate individuals’ current status and guide them toward a better direction.

However, we note that every individual is unique, varying in basic characteristics such as age, gender, and race, to more nuanced aspects like intellectual capacity and social status [26]; even a group of same-aged preschoolers exhibit large individual uniqueness [14, 15]. This is why a single standardized guideline does not universally apply to everyone; we inevitably find some aspects of standard guidelines that do not suit one’s unique circumstances. In other words, a standardized guideline is not a one-size-fits-all solution.

It is also not a practical solution for the experts to analyze individuals one by one to provide guidelines to each individual. For practicality, researchers analyze data from a large population to establish statistically representative norms and guidelines, deriving

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
UbiComp/ISWC ’23 Adjunct, October 08–12, 2023, Cancun, Quintana Roo, Mexico
© 2023 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 979-8-4007-0200-6/23/10...\$15.00
<https://doi.org/10.1145/3594739.3610761>

an average to construct a *standard persona*. Once such a norm is established, it often remains constant due to the enormous time and resources for renewing. In effect, standard guidelines often last for a long time, usually decades, yet are widely exercised by many, various people, in various times and circumstances.

Taken together, there are inherent limitations to these common norms. First, they may not be suitable for every individual, as such norms are inherently meant to be a single centroid approximating a large population of diverse individuals. This approach creates a discrepancy between common norms and one's own unique characteristics and social constraints. For instance, the recommended intake of 2000 calories per day may not be appropriate for some individuals, depending on their metabolic capacities [13] or daily activity volumes [27]. Second, applying the common norms to each individual's real life can be challenging. Even if these norms were ideal, one's real-life constraints may not allow one to follow the ideal norms [32]. For example, some people find it impractical to keep their bedtime early and consistent, due to late-night work obligations or irregular meetings with colleagues across time zones.

Given the limitations in applying common norms as-is, an individual often exercises workarounds: e.g., curating a subset of the guidelines that she can practically follow, adapting these guidelines to fit her own circumstances, or approximating herself to the persona that the standard norms presume. When all of these workarounds are not possible to follow, one may view herself as an outlier and disregard the norms. In the case of sleep, for instance, if you have a late-night meeting with overseas colleagues, or if your shift work necessitates you to be awake at dawn, adhering to standard sleep guidelines is not feasible. In such instances, we make a judgment call based on our own circumstances; we might go to bed early and wake up before the meeting, stay awake until the meeting and sleep after that, or if no other options work, stay up all night and catch up on sleep with a nap.

We address that the core issue with these common norms is that they present a standard model as a single entity, although the model is essentially a combined representation of the underlying *principles* and their *embodiment*. The underlying principle refers to the theory, empirical methods, and other foundations upon which the existing standard model is constructed, e.g., the circadian rhythm in sleep patterns [6]. Embodiment refers to the guidelines materialized based on these principles. It can be a guideline for “*when to actually go to bed and wake up*” based on the circadian rhythm, or “*which words this child should learn*” based on an age-appropriate language development benchmark. So far, it has been a common practice that the underlying principle has had many people's averaged data as an input, and the embodiment has existed as an averaged standard.

Given the issues identified with the existing standard model, our goal is to create personalized embodiment in context, while preserving the underlying principle. This requires technology on two fronts. The first is ubiquitous sensing technologies for personalized data collection in a practical manner. Rapidly advancing mobile computing and context inferencing technologies have been making it feasible to collect individual data from more and more facets of one's real-life [18, 19, 21, 23, 25, 29]. We will use such individual datasets as a foundation for personalization. The second one is a principled generation, which applies the collected data to

the principles to craft a new, dynamic, individual- and situation-specific embodiment. The extent and level of generation, along with the techniques to use, differ depending on the domain, requiring meticulous design in each case.

While many of the existing personalization works focus on curation [10, 17], we believe that the novelty of our work lies in the automatic *generation* of the embodiment. If curation is like fitting yourself into an off-the-rack suit, generation is like tailoring a custom suit. With curation, there is no assurance that any of the numerous existing guidelines will align with your circumstances, and even if you find the closest match, it may not be a perfect fit. Moreover, your circumstances change over time; hence you may need to undertake the curation process again. However, generation overcomes these limitations inherent to curation by harnessing the power of automation and personalization. We term this approach of overcoming the limitations of conventional personalized methods through principled generation as *hyper-personalization*.

We aim to design a framework that is adaptable to a wide range of potential domains. As an initial step, we have identified the efficacy of this approach within the realm of sleep. Most people have a good understanding of when and how much they should sleep for a healthy sleep life. Nonetheless, numerous real-life constraints make it challenging to put these guidelines into practice. We designed and implemented SleepGuru [24], a system that recommends actionable sleep schedules by considering an individual's real-life constraints. We detail SleepGuru in §2. Another area we are going to investigate is enhancing children's vocabulary learning and assessment, which is outlined in §3. §4 presents our future work to construct the overall framework, and §5 suggests the expected contribution of our work.

2 RESEARCH DONE : SLEEP RECOMMENDATION

2.1 Motivation

The motivation for the personalized sleep recommendation work [24] stemmed from the disconnection between existing standard sleep guidelines and the realities of our daily lives. With the advancement of personal devices, mobile & IoT systems, and computational sleep models, we can now track sleep duration and stages with commercial apps like Fitbit [2] and Google Health [3], which offer guidance on improving sleep. However, these standard guidelines assume ideal conditions and have many limitations when they are put into practice. For example, it is impossible to follow the system's advice that “*go to bed around 10 pm to facilitate healthy sleep*,” if you have a conflicting compulsory reason, such as a scheduled meeting at night. Therefore, we decided to design a sleep planning system respecting users' real-life constraints.

There exists a large body of work focusing on pervasive feedback and recommendations to promote healthy sleep. One such kind of work promotes users' awareness of standard recommendations in their daily life. ShutEye [5] designs an interactive smartphone wallpaper, visualizing standard sleep recommendations and their significance along time-of-day, e.g., “*Caffeinated drinks negatively impact sleep quality at later hours*.” SleepTight [8] advances further by incorporating the user's logging of her own sleep-influential actions and facilitating self-reflection with a graphical summary.

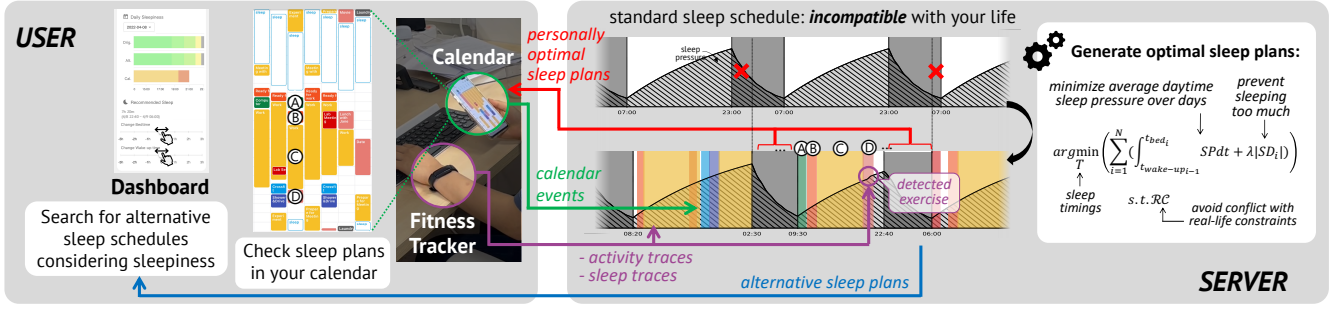


Figure 2: SleepGuru, a system for computing personally optimal sleep plans appreciating the user’s real-life constraints

A next body of works focus on helping users identify standard recommendations that matter in one’s lifestyle. SleepCoacher [10] facilitates informed curation by guiding a user through a series of self-experimentation with a pool of standard recommendations, e.g., “For the next 6 days, try going to bed regularly around 11 pm.”, so that she finds a subset of standard recommendations that help her sleep. Daskalova et al. [9] explores cohort-based sleep recommendations, in which users are grouped by similar physical profiles so that the same-group users share a set of selected recommendations.

However, all of these previous works have been *agnostic* to the user’s other real-life duty, requirements, and constraints that the user often prioritizes over sleep. Often, the user may find that the prompted recommendation conflicts with one of their real-life constraints, e.g., a late meeting or a night shift scheduled today. Now that the conflicting recommendation is hardly executable; she could either discard it or improvise on her own. To make things more complicated, such constraints vary day by day, making today’s improvisation inapplicable tomorrow. Furthermore, a one-day deviation from the standard may produce carry-over effects in the coming days, enlarging the gap between the user’s real-life and what is assumed in standard recommendations.

What we need is a sleep recommendation system that does not separate the user’s sleep from what is important in their real-life, which led to the development of SleepGuru – a system to dynamically *generate* personalized sleep plans such that the recommended plan is actionable within each individual’s real-life constraints.

2.2 Methodology

While building the personalized sleep recommendation system, we proceeded with the following steps. First, we uncovered the underlying principles behind standard sleep models. The literature on sleep physiology teaches the foundational principles of how our sleepiness develop, based on which the well-known standard bedtime and wake-up times are derived [28]. We adopt a metric called sleep pressure, which combines our diurnal circadian rhythms and homeostatic sleep drive. The standard sleep schedule means the times regulating one’s sleep pressure under a natural circadian rhythm, going to bed when the sleep pressure gets high, and waking up when the sleep pressure is relieved after enough sleep.

Second, we developed a model that treats the sleep guidelines not as a fixed embodiment but as on-demand computable ones. By doing so, our system can dynamically generate sleep guidelines

based on one’s sleep pressure prediction and real-life constraints, so that people can take them into practice in a way that makes sense in their real-life. We created an objective function for optimization to determine bedtime and wake-up times that minimize sleepiness during one’s wake period. We integrated real-world constraints into the optimization, categorizing them as hard and soft constraints. Although numerous factors affect one’s sleep, we focused on the physical state and time constraints in this work, as they are dynamic factors that can change multiple times a day, which highly benefit from ubiquitous sensing technologies.

Lastly, we implemented a system that generates a dynamic, personalized sleep plan that is easy to follow. Whenever the user’s constraints change, a new sleep plan is automatically computed and provided to the user. We also developed a dashboard interface to provide alternatives if the user is unable to follow the initial plan.

2.3 Evaluation

For the main deployment study, we deployed the SleepGuru system to 20 users for 8 weeks, and identified that the use of SleepGuru involved positive changes in terms of various aspects. The most prominent ones were user-perceived sleep quality and their rate of compliance to the recommendations, compared against both their self-scheduling (i.e., observation phase) and clinically desired scheduling but not mindful of their real-life (i.e., standard phase). The sleep quality measured with Leeds Sleep Evaluation Questionnaire (LSEQ) improved 15% over the observation phase, and 11% over the standard phase. The main reason for this improvement in sleep quality was that the SleepGuru system made followable recommendations. The compliance rate, a ratio that the participants followed the recommended sleep timings, was nearly three times higher in the SleepGuru phase (46%) compared to the standard phase (16%). Thus, the SleepGuru system effectively addressed the fundamental issue that the current standardized norms aimed to resolve, in both reasonable and practical manner.

3 RESEARCH ON-GOING : CHILDREN’S LANGUAGE LEARNING

3.1 Motivation

Language is a critical tool for communication. Newborns initially have no knowledge of the language, but gradually acquires it through social interactions with parents and exposure to external language stimuli. Missing the right time for language development can have

a negative impact on a child's later development such as social skills [31] and scholastic performance [7]. Thus, it is important to assess whether the child's language development is on track or not, and if necessary, to bring the child to timely and appropriate intervention from an SLP (Speech-Language Pathologist).

HCI researchers have explored ways to provide language guidance in an automated manner. TalkBetter [16, 17] tracks dialogue exchanges between parents and children, offering reminders for parents of the parent training guidelines. Similarly, TalkLIME [30] delivers feedback via monitoring the interactions between children and parents, employing phone screens as a modality to provide immediate responses. Captivate [22] is a contextual advisory system designed to assist parents in figuring out the most effective words to teach in the play situation. However, previous studies have focused on the instant moment of language learning, providing limited feedback, and have not comprehensively examined the language learning process: from assessing a child's current development stage to delivering suitable interventions.

The standard cycle of vocabulary learning is divided into two phases: assessment and intervention. Assessment means evaluating the child's current level, usually comparing their performance with peers of similar ages. SLPs use standard tools such as M-B CDI [12]. Intervention is providing therapeutic and/or pedagogical inputs for the child. Reading books [11, 20] is one of the most common approaches in intervention. The existing assessment and intervention cycles possess limitations in personalization, automation, and routinization. The standard tools are expensive to create and modify, making it difficult to adapt them to the evolving times and specific language environment. Therefore, these tools have limitations in assessing a child's language skills fairly. Identifying the target vocabulary that a child needs to learn, and teaching them in a suitable manner, are also challenges that call for personalization. Thus, we aim to develop a system employing a computational approach to personalize and automate this cycle. As a first step, we are trying to construct a personalized vocabulary set which can be used to assess each child's language development level.

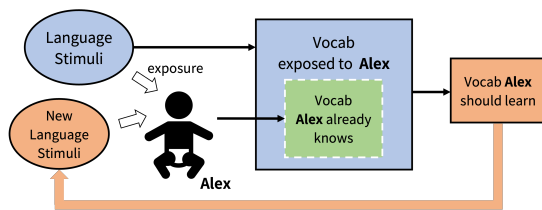


Figure 3: current practice in children language learning

3.2 Methodology

By monitoring the ever-present stimuli throughout a child's daily life, we can figure out the vocabulary they need to know. Through the speech that the child makes and hears, we can also identify the words they likely already know. We can infer the target words that the child should learn with these two sets of words. By leveraging rapidly advancing large language models and image-generative AI, we can create and provide customized picture books for children, adapted to the words the child should learn.

4 RESEARCH PLANNED : FRAMEWORK

We believe that our approach is not limited to a single domain. Through the case studies in various domains, we aim to identify the discrepancies between common norms and individual practices, and accordingly, devise a tailored principled generation strategy and prototype system for each. Subsequently, leveraging insights garnered from these diverse studies, we anticipate that we can categorize areas where our approach can be implemented based on the defined criteria, and eventually develop a formative framework that could be applied to potential cases in the future.

As an initial step, we propose that future research on principled generation can be carried out with the following methodology. While most of these suggestions stem from insights gained from sleep and language learning domain, we address that the methodology can be universally applied across different fields. First, we need to distinguish the underlying principles from the standard guidelines. To personalize the embodiment, we need to find out the fundamental principles shaping the common norms. That is, it is necessary to figure out how the standard guidelines were created. Second, we need to develop a method for personalizing the implementation based on an individual's own constraints and requirements. This is where cutting-edge ubiquitous sensing technologies and context-aware technologies come into play, allowing us to identify the real-life constraints of an one's lifestyle. Third, the personalized embodiment needs be generated and provided to the individual in a way that makes sense within each individual's real-life while still abiding by the underlying principles. Here, we should devise a method for generating this personalized embodiment automatically and dynamically. Given that each person possesses their own tendencies and preferences, it is essential to provide a personalized delivery method.

The presence of generation could be considered the key differentiator between the conventional personalization approach and the hyper-personalization we plan to realize.

5 EXPECTED CONTRIBUTION

We envision that the case-studies and framework, that have been or will be presented in this dissertation, would derive high-level lessons not only for the respective domains but also to shed light on the next-step issues of diversity and inclusion. We aim to identify and bridge the existing gap between principle and action, which is currently overlooked in the present standardized systems. Our objective is to create embodiments that are not only worth following but are also practical and feasible to follow.

In particular, this research could be a solution to embrace the diversity issues that have recently emerged. Despite diversity being a significant topic in a number of fields, it is still challenging to find a system that practically embraces it. More of the time, individuals representing diverse backgrounds and traits are left to seek out tailored services themselves, investing their own time and resources. Therefore, we expect to contribute to the field of HCI and ubiquitous computing as a generalized framework that can be implemented by individuals in various fields.

BIOGRAPHICAL SKETCH

Jungeun Lee is a Ph.D. student at Pohang University of Science and Technology (POSTECH), advised by Prof. Inseok Hwang. She began her Ph.D. program in September 2021 and is expected to graduate in 2025. Her research interest lies in designing personalization systems which respect individual's diversities and real-life constraints. She has published her initial work on personalized sleep recommendations at ACM UIST 2022.

ACKNOWLEDGMENTS

This research was supported by the National Research Foundation of Korea(NRF) grant funded by the Korea government(MSIT) (No. 2021R1A2C200386612). This research was also partly supported by the IITP grants by the MSIT, Korea (IITP-2020-0-01778), by the Alchemist Project NTIS1415187366(20025752) funded by the Ministry of Trade, Industry & Energy(MOTIE, Korea), and by the KOCCA grant funded by the Ministry of Culture, Sports and Tourism in 2022 (Project Number: R2021040136, Contribution Rate: 20%).

REFERENCES

- [1] [n. d.]. 2020 – 2025 Dietary Guidelines for Americans. Retrieved June. 7, 20 from https://www.dietaryguidelines.gov/sites/default/files/2021-03/Dietary_Guidelines_for_Americans-2020-2025.pdf
- [2] [n. d.]. Fitbit. Retrieved Jun. 9, 2023 from <https://www.fitbit.com>
- [3] [n. d.]. Google Health. Retrieved June. 9, 20 from <https://www.health.google>
- [4] [n. d.]. Typical Speech and Language Development. Retrieved June. 7, 20 from <https://www.asha.org/public/speech/development/>
- [5] Jared S Bauer, Sunny Consolvo, Benjamin Greenstein, Jonathan Schooler, Eric Wu, Nathaniel F Watson, and Julie Kientz. 2012. ShutEye: encouraging awareness of healthy sleep recommendations with a mobile, peripheral display. In *Proceedings of the SIGCHI conference on Human factors in computing systems*. Association for Computing Machinery, New York, NY, USA, 1401–1410.
- [6] Christian Cajochen, K Kräuchi, and A Wirz-Justice. 2003. Role of melatonin in the regulation of human circadian rhythms and sleep. *Journal of neuroendocrinology* 15, 4 (2003), 432–437.
- [7] Hugh W Catts. 1997. The early identification of language-based reading disabilities. *Language, Speech, and Hearing Services in Schools* 28, 1 (1997), 86–89.
- [8] Eun Kyoung Choe, Bongshin Lee, Matthew Kay, Wanda Pratt, and Julie A Kientz. 2015. SleepTight: low-burden, self-monitoring technology for capturing and reflecting on sleep behaviors. In *Proceedings of the 2015 ACM International Joint Conference on Pervasive and Ubiquitous Computing*. Association for Computing Machinery, New York, NY, USA, 121–132.
- [9] Nediya Daskalova, Bongshin Lee, Jeff Huang, Chester Ni, and Jessica Lundin. 2018. Investigating the effectiveness of cohort-based sleep recommendations. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 2, 3 (2018), 1–19.
- [10] Nediya Daskalova, Danaë Metaxa-Kakavouli, Adrienne Tran, Nicole Nugent, Julie Boergers, John McGeary, and Jeff Huang. 2016. SleepCoach: A personalized automated self-experimentation system for sleep recommendations. In *Proceedings of the 29th annual symposium on user interface software and technology*. Association for Computing Machinery, New York, NY, USA, 347–358.
- [11] Nicholas Dowdall, Gerardo J Melendez-Torres, Lynne Murray, Frances Gardner, Leila Hartford, and Peter J Cooper. 2020. Shared picture book reading interventions for child language development: A systematic review and meta-analysis. *Child development* 91, 2 (2020), e383–e399.
- [12] Larry Fenson et al. 2007. MacArthur-Bates communicative development inventories. (2007).
- [13] J Arthur Harris and Francis G Benedict. 1918. A biometric study of human basal metabolism. *Proceedings of the National Academy of Sciences* 4, 12 (1918), 370–373.
- [14] Inseok Hwang, Hyukjae Jang, Lama Nachman, and Junehwa Song. 2010. Exploring inter-child behavioral relativity in a shared social environment: a field study in a kindergarten. In *Proceedings of the 12th ACM international conference on Ubiquitous computing*. 271–280.
- [15] Inseok Hwang, Hyukjae Jang, Taiwoo Park, Aram Choi, Youngki Lee, Chanyou Hwang, Yangui Choi, Lama Nachman, and Junehwa Song. 2012. Leveraging children's behavioral distribution and singularities in new interactive environments: Study in kindergarten field trips. In *Pervasive Computing: 10th International Conference, Pervasive 2012, Newcastle, UK, June 18–22, 2012. Proceedings 10*. Springer, 39–56.
- [16] Inseok Hwang, Youngki Lee, Chungkuk Yoo, Chulhong Min, Dongsun Yim, and John Kim. 2019. Towards interpersonal assistants: next-generation conversational agents. *IEEE Pervasive Computing* 18, 2 (2019), 21–31.
- [17] Inseok Hwang, Chungkuk Yoo, Chanyou Hwang, Dongsun Yim, Youngki Lee, Chulhong Min, John Kim, and Junehwa Song. 2014. TalkBetter: family-driven mobile intervention care for children with language delay. In *Proceedings of the 17th ACM conference on Computer supported cooperative work & social computing*. 1283–1296.
- [18] Bumsoo Kang, Inseok Hwang, Jinho Lee, Seungchul Lee, Taegyeong Lee, Youngjae Chang, and Min Kyung Lee. 2018. My being to your place, your being to my place: Co-present robotic avatars create illusion of living together. In *Proceedings of the 16th Annual International Conference on Mobile Systems, Applications, and Services*. 54–67.
- [19] Bumsoo Kang, Seungwoo Kang, and Inseok Hwang. 2021. Momentmeld: Ai-augmented mobile photographic memento towards mutually stimulatory intergenerational interaction. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–16.
- [20] Bumsoo Kang, Chulhong Min, Wonjung Kim, Inseok Hwang, Chunjong Park, Seungchul Lee, Sung-Ju Lee, and Junehwa Song. 2017. Zaturi: We put together the 25th hour for you. create a book for your baby. In *Proceedings of the 2017 ACM Conference on Computer Supported Cooperative Work and Social Computing*. 1850–1863.
- [21] Wonjung Kim, Seungchul Lee, Seonghoon Kim, Sunbin Jo, Chungkuk Yoo, Inseok Hwang, Seungwoo Kang, and Junehwa Song. 2020. Dyadic Mirror: Everyday Second-person Live-view for Empathetic Reflection upon Parent-child Interaction. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 4, 3 (2020), 1–29.
- [22] Taeahn Kwon, Minkyung Jeong, Eon-Suk Ko, and Youngki Lee. 2022. Captivate! contextual language guidance for parent–child interaction. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–17.
- [23] Haechan Lee, Miri Moon, Taiwoo Park, Inseok Hwang, Uichin Lee, and Junehwa Song. 2013. Dungeons & swimmers: designing an interactive exergame for swimming. In *Proceedings of the 2013 ACM conference on Pervasive and Ubiquitous Computing adjunct publication*. 287–290.
- [24] Jungeun Lee, Sungnam Kim, Minki Cheon, Hyojin Ju, JaeEun Lee, and Inseok Hwang. 2022. SleepGuru: Personalized Sleep Planning System for Real-life Actionability and Negotiability. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology*. 1–16.
- [25] Youngki Lee, Younghyun Ju, Chulhong Min, Seungwoo Kang, Inseok Hwang, and Junehwa Song. 2012. Comon: Cooperative ambience monitoring platform with continuity and benefit awareness. In *Proceedings of the 10th international conference on Mobile systems, applications, and services*. 43–56.
- [26] Marilyn Loden. 1996. *Implementing diversity*. Irwin Professional Chicago, IL.
- [27] Chulhong Min, Chungkuk Yoo, Inseok Hwang, Seungwoo Kang, Youngki Lee, Seungchul Lee, Pillsoon Park, Changhun Lee, Seungpyo Choi, and Junehwa Song. 2015. Sandra helps you learn: The more you walk, the more battery your phone drains. In *Proceedings of the 2015 ACM international joint conference on pervasive and ubiquitous computing*. 421–432.
- [28] Andrew JK Phillips, Elizabeth B Klerman, and James P Butler. 2017. Modeling the adenosine system as a modulator of cognitive performance and sleep patterns during sleep restriction and recovery. *PLoS computational biology* 13, 10 (2017), e1005759.
- [29] Meera Radhakrishnan, Darshana Rathnayake, Ong Koon Han, Inseok Hwang, and Archan Misra. 2020. ERICA: enabling real-time mistake detection & corrective feedback for free-weights exercises. In *Proceedings of the 18th Conference on Embedded Networked Sensor Systems*. 558–571.
- [30] Seokwoo Song, Seunggho Kim, John Kim, Wonjeong Park, and Dongsun Yim. 2016. TalkLIME: mobile system intervention to improve parent-child interaction for children with language delay. In *Proceedings of the 2016 ACM International Joint Conference on Pervasive and Ubiquitous Computing*. 304–315.
- [31] Susan E Stothard, Margaret J Snowling, Dorothy VM Bishop, Barry B Chipchase, and Carole A Kaplan. 1998. Language-impaired preschoolers: A follow-up into adolescence. *Journal of Speech, Language, and Hearing Research* 41, 2 (1998), 407–418.
- [32] Patricia M Wong, Brant P Hasler, Thomas W Kamarck, Matthew F Muldoon, and Stephen B Manuck. 2015. Social jetlag, chronotype, and cardiometabolic risk. *The Journal of Clinical Endocrinology & Metabolism* 100, 12 (2015), 4612–4620.